L6

Supply Chain · Recommendations · Pricing

Indian retail scale makes AI the infrastructure, not a feature. Swiggy 20 lakh orders/day. Nykaa 3 cr users. At this scale, traditional operations break; AI is the plumbing.

Which Indian retail app personalises most for you?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

Flipkart BBD 2024: 9 crore products in 5 days. No human in the loop.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. 4 Indian retail AI applications.
2. BBD-scale demand forecasting.
3. Recommendation engines.
4. Dynamic pricing strategies.
5. Returns-prediction model.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
4 applications
Key concept of this lesson.

02
BBD forecasting design
LightGBM at 8 crore pairs

03
Recommendations
Layered approach

04
Returns prediction
Pre + post order

01

PART 1 OF 4

4 applications

Key concept of this lesson.

CONCEPT 1 OF 4

4 applications

A core idea you'll use throughout this lesson.

DEFINITION

4 applications

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

BBD forecasting design

Features: lag, rolling, seasonality, promo.
Hierarchical LightGBM per category.
Refresh hourly during BBD.
MAPE target < 18%.
WHY NOT DL
Tabular

LightGBM wins 95% of tabular retail.

KEY POINTS

- Features: lag, rolling, seasonality, promo
- Hierarchical LightGBM per category
- Refresh hourly during BBD
- MAPE target < 18%

CONCEPT 3 OF 4

Recommendations

Collaborative filter for cold start.
Two-tower NN (user, item).
CLIP for visual similarity.
Contextual bandits per session.
A/B test every change.
HABIT
A/B always

Never deploy blind.

DEFINITION

Recommendations

Layered approach

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Returns prediction

Signal: fit, colour, description.
Pre-order: warn on > 30% prob.
Post-order: prioritise low-return in fulfilment.
Myntra: ₹120 cr/yr saved.
IMPACT
₹120 cr/yr

Size recommendation alone.

KEY POINTS

- Signal: fit, colour, description
- Pre-order: warn on > 30% prob
- Post-order: prioritise low-return in fulfilment
- Myntra: ₹120 cr/yr saved

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.
2.
3.
4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1	*Where will you apply this in your next project?*
Q2	*What was the single biggest 'aha' from this lesson?*
Q3	*Which habit will you start using this week?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Why LightGBM over DL for retail:

- **A** Speed
- **B** Tabular + interpretable + fast retrain
- **C** Size
- **D** None

ANSWER · QUESTION 1 OF 3

Why LightGBM over DL for retail:

B

CORRECT ANSWER

Tabular + interpretable + fast retrain

WHY

95% of tabular wins

KNOWLEDGE CHECK · QUESTION 2 OF 3

Myntra size recommendation impact:

A Nothing

B ₹120 cr/yr

C ₹1 L

D Random

ANSWER · QUESTION 2 OF 3

Myntra size recommendation impact:

CORRECT ANSWER

B

₹120 cr/yr

WHY

Return-rate reduction

KNOWLEDGE CHECK · QUESTION 3 OF 3

Two-tower embeds:

A None

B User + item

C Price + discount

D Image + text

ANSWER · QUESTION 3 OF 3

Two-tower embeds:

CORRECT ANSWER

B

User + item

WHY

Dot product scores

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

1. 4 applications
2. BBD forecasting design
3. Recommendations
4. Returns prediction

APPLY IT → ASSIGNMENT

Indian retail is the scale lab.

"Everything else is a prototype."

DELIVERABLES

- NEXT
- Lesson 7 — Climate.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

> *Knowing the right tool is half the battle. Doing the small thing today is the other half.*

— AI Learning Hub

END OF LESSON 9

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.